

CO2 ASSESSMENT TOOL 3 DEBUGGING & OPTIMISATION TASK

Aim

Debug brute force Convex Hull logic for collinear cases.

Bug Explanation

The original code rejects every collinear point ($\text{orientation} == 0$). Hull edges may legitimately contain collinear boundary points. Only points lying outside the candidate edge should invalidate it.

Corrected Algorithm

Determine orientation for all points. Accept collinear points. Reject only if points exist on both sides. For collinear points ensure they lie on the segment.

Improved Codelike Python

```
def orientation(a,b,c):
    return (b[0]-a[0])*(c[1]-a[1])-(b[1]-a[1])*(c[0]-a[0])

def on_segment(a,b,p):
    return min(a[0],b[0])<=p[0]<=max(a[0],b[0]) and min(a[1],b[1])<=p[1]<=max(a[1],b[1])

def is_hull_edge(p1,p2,points):
    side=0
    for p in points:
        if p==p1 or p==p2:
            continue
        val=orientation(p1,p2,p)
        if val==0:
            if not on_segment(p1,p2,p):
                return False
            continue
        cur=1 if val>0 else -1
        if side==0:
            side=cur
        elif side!=cur:
            return False
    return True
```

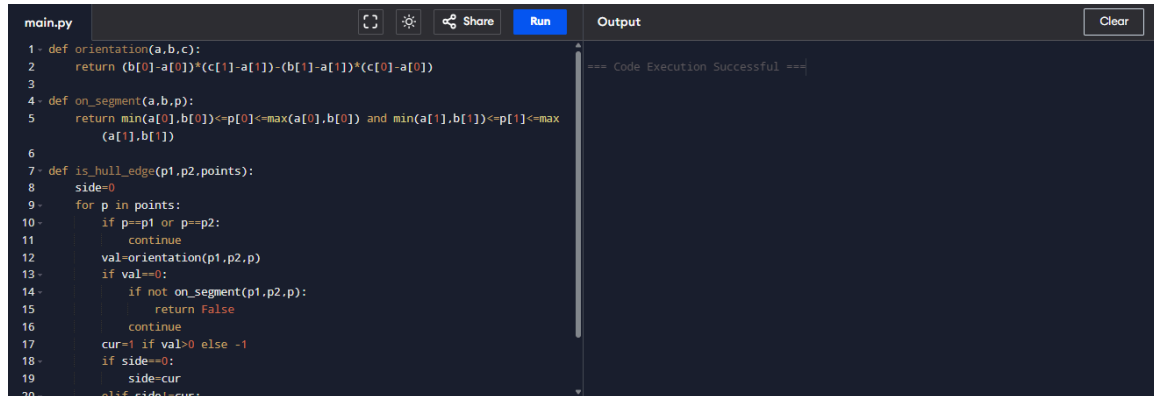
Optimization

Only one orientation computation per point. Side tracking avoids repeated comparisons.

Time Complexity

Brute Force: $O(n^3)$. Orientation checks are $O(1)$.

Output Screenshot



```
main.py [Icons] [Share] [Run] [Clear]
1- def orientation(a,b,c):
2-     return (b[0]-a[0])*(c[1]-a[1])-(b[1]-a[1])*(c[0]-a[0])
3-
4- def on_segment(a,b,p):
5-     return min(a[0],b[0])<=p[0]<=max(a[0],b[0]) and min(a[1],b[1])<=p[1]<=max(a[1],b[1])
6-
7- def is_hull_edge(p1,p2,points):
8-     side=0
9-     for p in points:
10-         if p==p1 or p==p2:
11-             continue
12-         val=orientation(p1,p2,p)
13-         if val==0:
14-             if not on_segment(p1,p2,p):
15-                 return False
16-             continue
17-         cur=1 if val>0 else -1
18-         if side==0:
19-             side=cur
20-         elif side!=cur:
```

=== Code Execution Successful ===

Conclusion

Correct handling of collinear boundary points ensures valid hull edges while preserving brute-force complexity.